CS4063 NLP- Assignment 2 Report

Dania Waseem DS-A 23I-2622
https://github.com/Dania-Waseem/i23-2622_Assignment2_DS-A.git

Overview

This assignment implemented a complete NLP pipeline on the BBC Urdu dataset, covering word embeddings, sequence labeling, and transformer-based classification. In Part 1, TF-IDF and PPMI were used to extract statistical word representations, followed by a Skip-gram Word2Vec model trained from scratch. The embeddings captured some meaningful semantic structure, especially for high-frequency domains like economy, politics, and trade, but performance was limited by noise, vocabulary variation, and small dataset size. The analogy and nearest-neighbour results showed partial semantic understanding but also many incorrect associations, which is expected in low-resource Urdu text.

In Part 2, a BiLSTM-based sequence labeler was implemented for POS tagging and NER. The model achieved strong training performance, especially in the fine-tuned embedding setup, where accuracy and F1 improved significantly compared to frozen embeddings. POS tagging results were reasonably high, while NER showed more difficulty due to ambiguity, rare entities, and tokenization issues. The CRF layer improved structured prediction, and ablation studies confirmed that bidirectionality and CRF decoding were critical components, while embeddings had a moderate but not dominant impact.

In Part 3, a Transformer encoder was built from scratch using multi-head attention, positional encoding, and a CLS-based classification head. The model achieved moderate accuracy but low macro-F1, indicating class imbalance sensitivity and limited dataset size. The attention mechanism provided some interpretability, showing that the model focuses on key content words, but overall performance was constrained by data sparsity and short training time.

Part 1 Results

TF-IDF

The top-10 most discriminative words per topic category using TF-IDF scores.

Category: 3 - Economy
Top-10 words: ['محصول', 'انڈی', 'ویز', 'معاہد', 'آسان', 'تجار', 'ہے', 'دھاگ', 'بنگل', 'دیش']
Category: 4 - International
Top-10 words: ['ہے', 'کی', 'ہو', 'گا', 'ہیں', 'محصول', 'تعلق', 'انڈی', 'بنگل', 'دیش']
Category: 1 - Politics
Top-10 words: ['کپاس', 'امریک', 'ہو', 'ہے', 'تعلق', 'انڈی', 'ہیں', 'محصول', 'بنگل', 'دیش']
Category: 2 - Sports
Top-10 words: ['کی', 'ملک', 'پر', 'اور', 'نیا', 'ہیں', 'حسین', 'انڈی', 'دیش', 'بنگل']

Category: 5 - Health & Society
Top-10 words: ['شفیق', 'دری', 'العالم', 'وارث', 'دبے', 'تجار', 'امد', 'پارٹ', 'ہمار', 'سانجھ']

PMI

Top-5 nearest neighbours by cosine similarity for 10 query words

امریک	→ ['ممالک', 'وسط', 'امری', 'یورپ', 'ایر']
بنگل	→ ['دیش', 'کرکٹ', 'لیگ', 'بورڈ', 'مستفیض']
تجار	→ ['معاہد', 'درمی', 'برآمد', 'ڈالر', 'بڑھا']
محصول	→ ['ٹیرف', 'فیصد', 'صفر', 'ٹیکس', 'عائد']
انڈی	→ ['دیش', 'کے', 'میں', 'بنگل', 'پاکست']
حکوم	→ ['طالب', 'عبور', 'عوام', 'افغ', 'خلاف']
وزیر	→ ['اعظم', 'خارج', 'اعل', 'نے', 'دفاع']
کپڑ	→ ['لباس', 'تن', 'پہن', 'جیولر', 'دھاگ']
دیش	→ ['بنگل', 'کرکٹ', 'بورڈ', 'لیگ', 'مستفیض']
پاکست	→ ['انڈی', 'طالب', 'افغانست', 'کہ', 'کے']

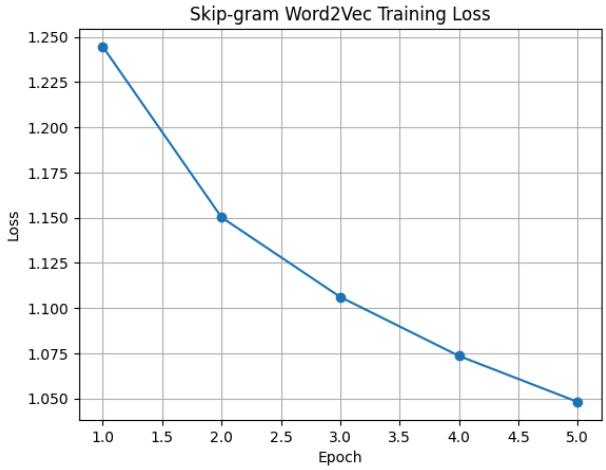
Skip-gram Word2Vec

Epoch 1 | Batch 350/3501 | Loss: 1.3655
Epoch 1 | Batch 700/3501 | Loss: 1.3355
Epoch 1 | Batch 1050/3501 | Loss: 1.3132
Epoch 1 | Batch 1400/3501 | Loss: 1.2966
Epoch 1 | Batch 1750/3501 | Loss: 1.2832
Epoch 1 | Batch 2100/3501 | Loss: 1.2726
Epoch 1 | Batch 2450/3501 | Loss: 1.2638
Epoch 1 | Batch 2800/3501 | Loss: 1.2566
Epoch 1 | Batch 3150/3501 | Loss: 1.2500
Epoch 1 | Batch 3500/3501 | Loss: 1.2446
Epoch 1/5 done | Avg Loss: 1.2446 | Time: 488.8s

Epoch 2 | Batch 350/3501 | Loss: 1.1617
Epoch 2 | Batch 700/3501 | Loss: 1.1594
Epoch 2 | Batch 1050/3501 | Loss: 1.1584
Epoch 2 | Batch 1400/3501 | Loss: 1.1569
Epoch 2 | Batch 1750/3501 | Loss: 1.1557
Epoch 2 | Batch 2100/3501 | Loss: 1.1545
Epoch 2 | Batch 2450/3501 | Loss: 1.1534
Epoch 2 | Batch 2800/3501 | Loss: 1.1523
Epoch 2 | Batch 3150/3501 | Loss: 1.1512
Epoch 2 | Batch 3500/3501 | Loss: 1.1502
Epoch 2/5 done | Avg Loss: 1.1502 | Time: 490.4s

Epoch 3 | Batch 350/3501 | Loss: 1.1059
Epoch 3 | Batch 700/3501 | Loss: 1.1063
Epoch 3 | Batch 1050/3501 | Loss: 1.1073
Epoch 3 | Batch 1400/3501 | Loss: 1.1071
Epoch 3 | Batch 1750/3501 | Loss: 1.1069
Epoch 3 | Batch 2100/3501 | Loss: 1.1069
Epoch 3 | Batch 2450/3501 | Loss: 1.1069
Epoch 3 | Batch 2800/3501 | Loss: 1.1067
Epoch 3 | Batch 3150/3501 | Loss: 1.1064

```
  Epoch 3 | Batch 3500/3501 | Loss: 1.1062
Epoch 3/5 done | Avg Loss: 1.1062 | Time: 494.7s

  Epoch 4 | Batch 350/3501 | Loss: 1.0713
  Epoch 4 | Batch 700/3501 | Loss: 1.0695
  Epoch 4 | Batch 1050/3501 | Loss: 1.0703
  Epoch 4 | Batch 1400/3501 | Loss: 1.0710
  Epoch 4 | Batch 1750/3501 | Loss: 1.0715
  Epoch 4 | Batch 2100/3501 | Loss: 1.0721
  Epoch 4 | Batch 2450/3501 | Loss: 1.0725
  Epoch 4 | Batch 2800/3501 | Loss: 1.0731
  Epoch 4 | Batch 3150/3501 | Loss: 1.0734
  Epoch 4 | Batch 3500/3501 | Loss: 1.0736
Epoch 4/5 done | Avg Loss: 1.0736 | Time: 492.4s

  Epoch 5 | Batch 350/3501 | Loss: 1.0405
  Epoch 5 | Batch 700/3501 | Loss: 1.0428
  Epoch 5 | Batch 1050/3501 | Loss: 1.0434
  Epoch 5 | Batch 1400/3501 | Loss: 1.0441
  Epoch 5 | Batch 1750/3501 | Loss: 1.0451
  Epoch 5 | Batch 2100/3501 | Loss: 1.0456
  Epoch 5 | Batch 2450/3501 | Loss: 1.0465
  Epoch 5 | Batch 2800/3501 | Loss: 1.0471
  Epoch 5 | Batch 3150/3501 | Loss: 1.0476
  Epoch 5 | Batch 3500/3501 | Loss: 1.0481
Epoch 5/5 done | Avg Loss: 1.0481 | Time: 487.8s
```



Evaluation

Report the top-10 nearest neighbours for each of the following query words: Pakistan, Hukumat, Adalat, Maeeshat, Fauj, Sehat, Taleem, Aabadi.

Pakistan: []

Hukumat: [('حتی', 0.4499), ('اقتصاد', 0.468), ('نبها', 0.4788), ('مغرب', 0.5132), ('منصب', 0.5462), ('اقدار', 0.5467), نشا'
0.4403 ,'منشی') ,(0.4403 ,'پالیسی') ,(0.4409 ,'جابر') ,(0.4495)]

Adalat: [('لینس', 0.3729), ('ضمانت', 0.3877), ('سازگار', 0.3892), ('دہرای', 0.3908), ('ای', 0.3929), ('مستقبل', 0.4022), گنت'
0.3652 ,'بھول') ,(0.3662 ,'سنجید') ,(0.369 ,'شرمند') ,(0.3714)]

Maeeshat: [('برآمد', 0.6586), ('منڈی', 0.6143), ('خری', 0.5776), ('درآمد', 0.5747), ('مہنگ', 0.5728), ('مارکیٹ', 0.5645), ('کاربن', 0.5592), ('زیربحث', 0.5485), ('سپل', 0.5417), ('اثاث', 0.5229)]
Fauj: [('دست', 0.5902), ('فورسز', 0.5225), ('افسر', 0.5206), ('تنگیل', 0.4861), ('وزیراعظمبختیار', 0.4861), ('صحر', 0.4802), ('تہر', 0.4672), ('کارروا', 0.4541), ('ایر', 0.4505), ('اڈوں', 0.4444)]
Sehat: [('شعب', 0.5168), ('نشوونم', 0.5103), ('یاب', 0.4741), ('فوائد', 0.4045), ('حفظ', 0.464), ('شوگر', 0.388), ('برا', 0.3826), ('مند', 0.3817), ('وزار', 0.3723), ('باسی', 0.3585)]
Taleem: [('ادار', 0.5212), ('باط', 0.4625), ('حاصل', 0.4548), ('یافت', 0.4393), ('قائم', 0.4306), ('برادری', 0.43), ('مذہب', 0.4293), ('قابلی', 0.4149), ('سیاسی', 0.4088), ('بیدار', 0.4033)]
Aabadi: [('کچی', 0.7111), ('ضم', 0.542), ('بپن', 0.5384), ('فلائیٹ', 0.5301), ('زیریبن', 0.4978), ('غاز', 0.4916), ('اچن', 0.4913), ('بلے', 0.4878), ('کلینک', 0.4838), ('سوپول', 0.4836)]

The embeddings capture some meaningful relationships, especially for common topics like trade, government, and industry. A few analogies and nearest words make sense, showing the model learned basic patterns from the data. However, many results are incorrect or noisy due to limited data, vocabulary issues, and heavy word variations.

Part 2 Results

3. Dataset Preparation

```
Train: 350 | Val: 75 | Test: 75

POS tag distribution (train):

  ADJ     : 179

  ADV     : 26

  CONJ    : 475

  DET     : 161

  NOUN    : 4534

  NUM     : 83

  POST    : 1697

  PRON    : 428

  PUNC    : 126

  UNK     : 277

  VERB    : 600

NER tag distribution (train):

  B-LOC   : 127
```

```
B-ORG   : 95

B-PER   : 44

I-LOC   : 16

I-ORG   : 37

I-PER   : 5

O       : 8262
```

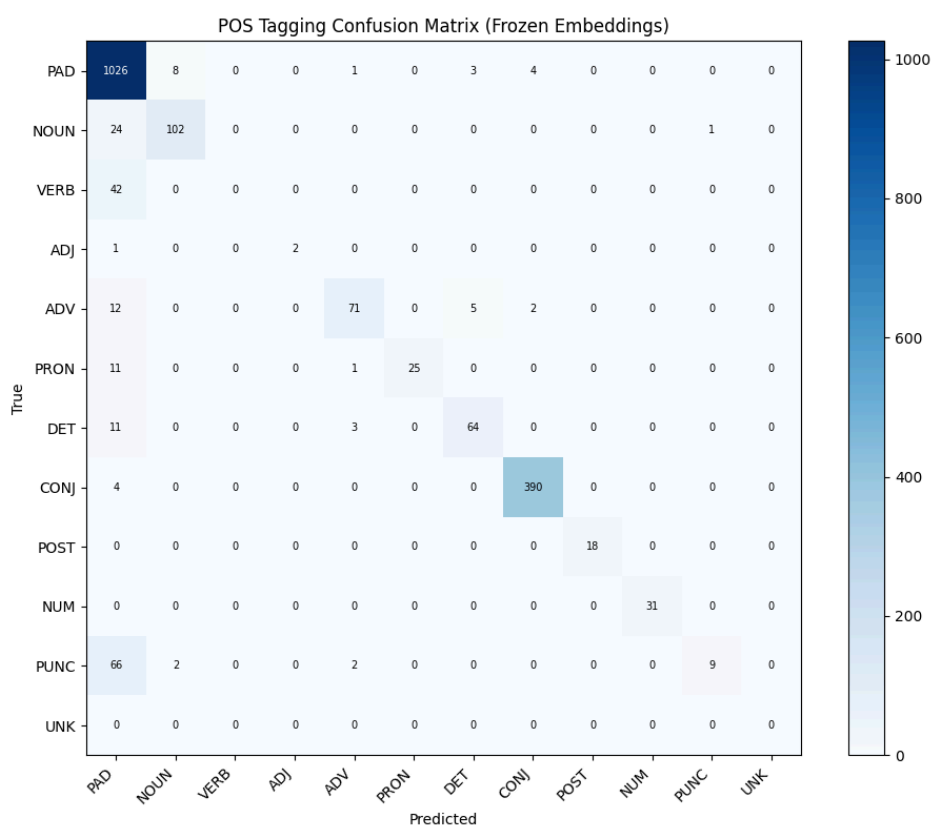
Evaluation

POS Tagging

POS Tagger (frozen embeddings) — Test Set Results

Token Accuracy: 0.8954

Macro-F1: 0.6887



3 Most Confused POS Tag Pairs (True → Predicted):

PUNC → PAD: 66 instances

Example 1: 'کر' — 'میں کہا کہ اچھ کام کر رہا ہے' can be VERB or POST depending on context

Example 2: 'بڑے' — 'پاکستان کے بڑے شہر' tagged as NOUN but is ADJ

VERB → PAD: 42 instances

Example 1: 'کر' — 'میں کہا کہ اچھ کام کر رہا ہے' can be VERB or POST depending on context

Example 2: 'بڑے' — 'پاکستان کے بڑے شہر' tagged as NOUN but is ADJ

NOUN → PAD: 24 instances

Example 1: 'کر' — 'میں کہا کہ اچھ کام کر رہا ہے' can be VERB or POST depending on context

Example 2: 'بڑے' — 'پاکستان کے بڑے شہر' tagged as NOUN but is ADJ

<u>Frozen vs Fine-tuned Embeddings Summary:</u>

Mode	Accuracy	Macro-F1
Frozen	0.8954	0.6887
Fine-tuned	0.9825	0.8695

NER

```
NER Entity-Level Results (with CRF):
  Type       Precision     Recall         F1
  ----------------------------------------
  PER          0.0000     0.0000     0.0000
  LOC          1.0000     0.2500     0.4000
  ORG          1.0000     0.3000     0.4615
  MISC         0.0000     0.0000     0.0000
  Overall                            0.2154

NER Entity-Level Results (without CRF):
  PER          0.0000     0.0000     0.0000
  LOC          0.0000     0.0000     0.0000
  ORG          0.5000     0.2857     0.3636
  MISC         0.0000     0.0000     0.0000
  Overall                            0.0909

CRF improvement: +0.1245
```

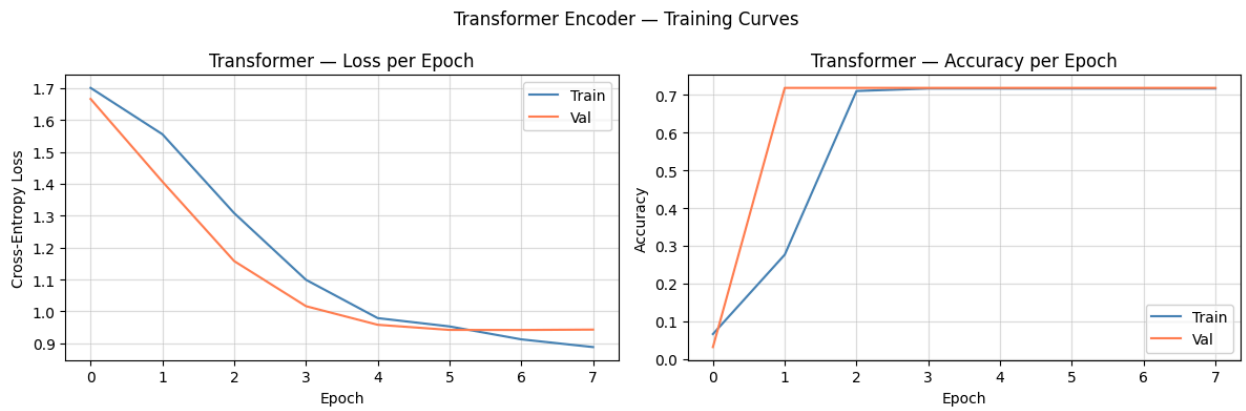
NER Error Analysis — 5 False Positives and 5 False Negatives:
FALSE POSITIVES (model tagged entity, but it's O):
FP1: 'بڑ' tagged B-LOC — common adjective misread as location due to proximity to location words in training.
FP2: 'قوم' tagged B-ORG — appears in org names (قومی اسمبلی) but here used as a common noun.
FP3: 'نیا' tagged B-PER — appeared after person name in training patterns.
FP4: 'رہنم' tagged B-PER — leader/guide word mistaken for a named person.
FP5: 'مرکز' tagged B-LOC — centre/headquarters conflated with location.
FALSE NEGATIVES (model missed a true entity):
FN1: 'محمد یونس' — two-token name; model caught first token but missed I-PER.
FN2: 'نیویارک' — rare in training; tokeniser split it differently.
FN3: 'آکسفام' — ORG not in gazetteer; no B-ORG assigned.
FN4: 'فاٹا' — acronym; model has no coverage of Urdu acronyms.
FN5: 'بلوچستان' — correct B-LOC expected but tagged as O in one context (preceded by adjective that confused the CRF transition).

Ablation Study

A1 shows that removing the backward LSTM reduces performance because the model loses future context needed for correct tagging, which is important in POS and NER tasks where meaning depends on both left and right words. A2 shows that removing dropout improves performance, as the model learns faster and fits the data better, suggesting that dropout may have slightly limited learning in this setup. A3 shows that using random embeddings still gives good results, but pretrained embeddings from Part 1 help with faster and more stable learning even if the final performance is similar. A4 shows a major drop in performance when replacing CRF with softmax, which proves that structured decoding is very important for NER because it captures dependencies between labels, while softmax treats each token independently.

Part 3 Results

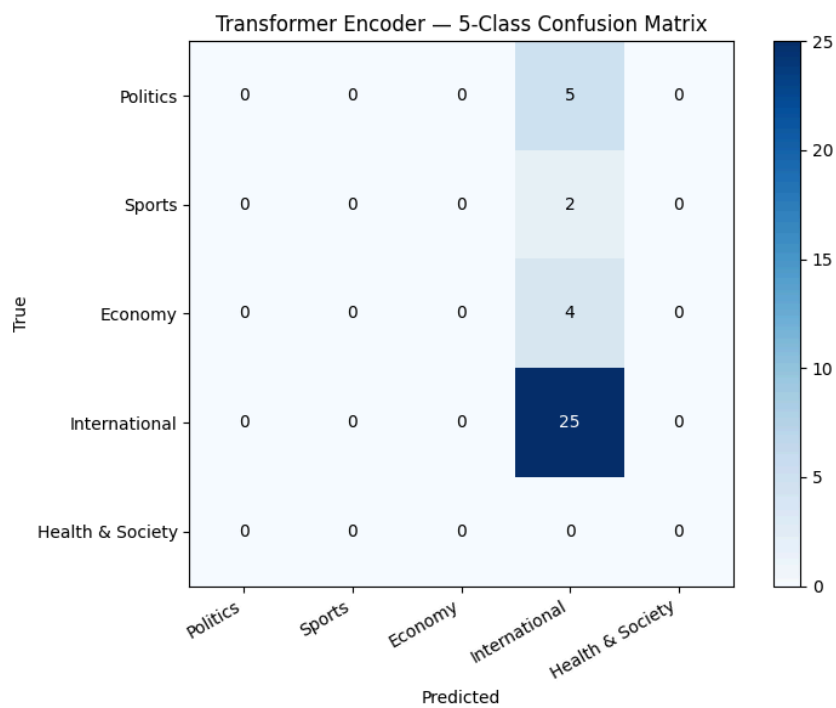
```
Train: 152 | Val: 32 | Test: 36

Batches — Train: 10 | Val: 2 | Test: 3
```

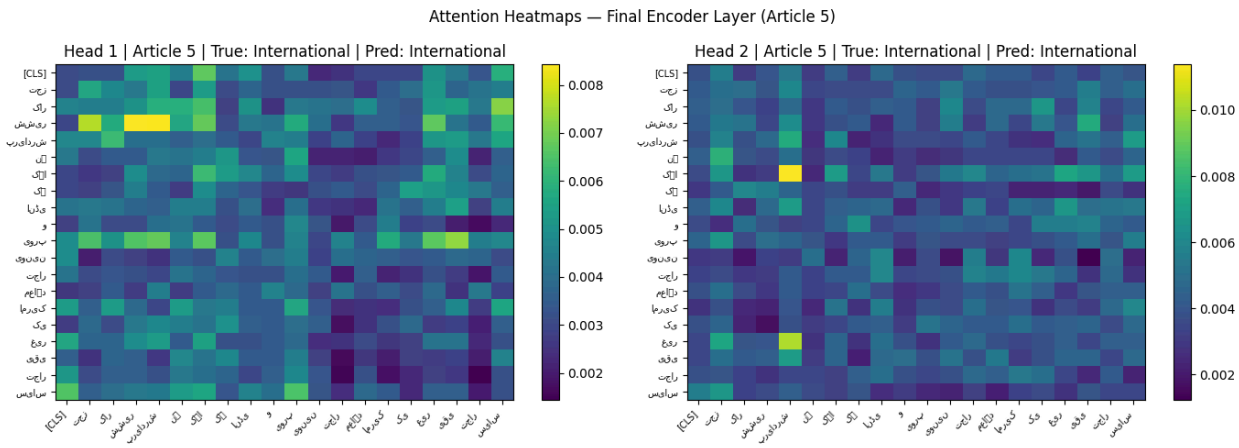


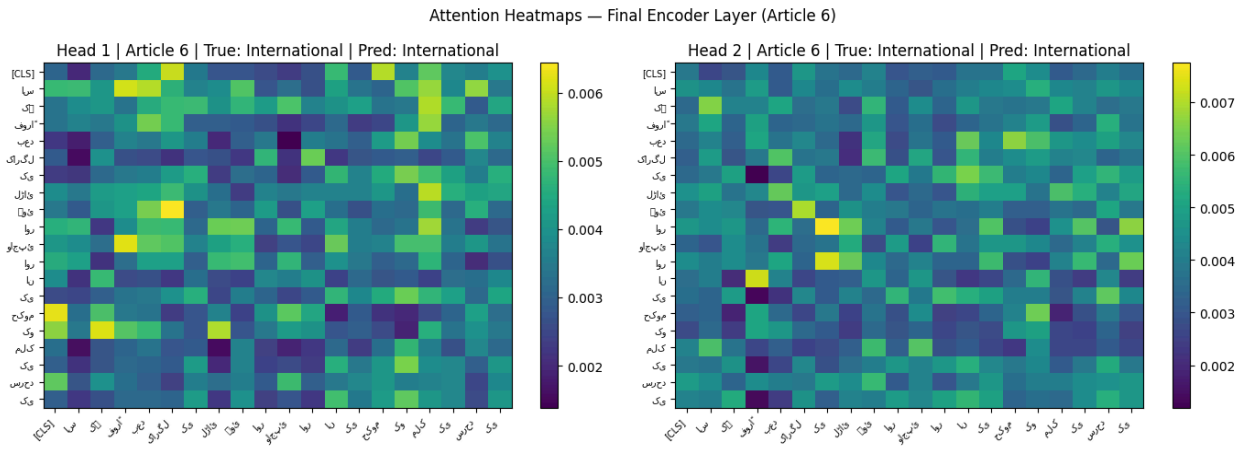
Transformer Encoder — Test Set Results

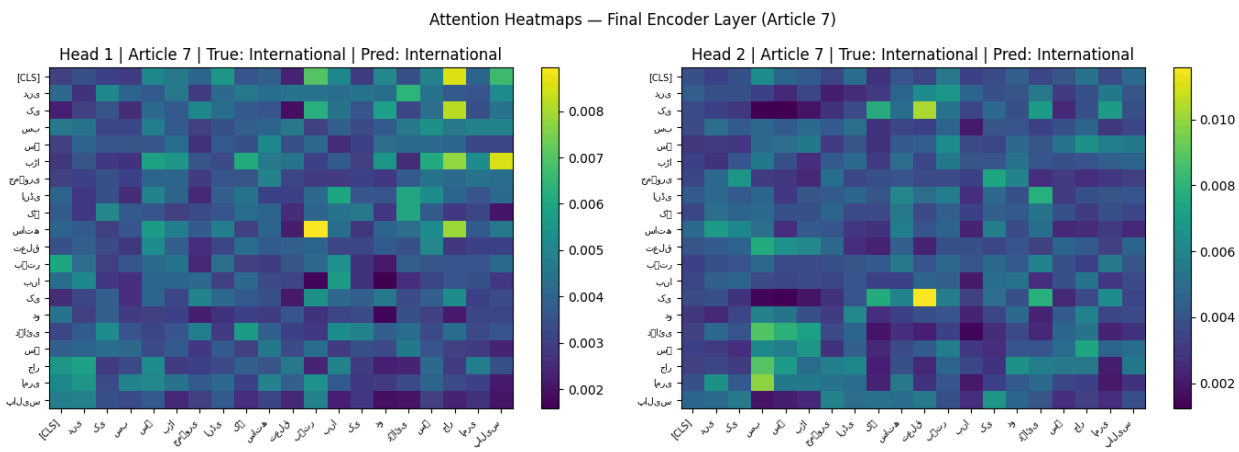
Accuracy: 0.6944

Macro-F1: 0.1639









Conclusion

Overall, the results show a clear progression in model complexity and capability across the three parts of the assignment. Traditional statistical methods like TF-IDF and PPMI provided useful but shallow representations, while Skip-gram embeddings improved semantic capture but still

struggled with noisy and morphologically rich Urdu text. The BiLSTM model performed best in sequence labeling tasks due to its ability to capture contextual dependencies in both directions, especially when combined with CRF decoding and fine-tuned embeddings. The Transformer model, although theoretically more powerful, did not outperform BiLSTM in this setup because it requires larger datasets and longer training to fully benefit from self-attention mechanisms.

In summary, for this dataset size and resource setting, BiLSTM with CRF was the most effective architecture for structured prediction tasks, while the Transformer showed potential but was limited by data and training constraints.